4pSCa19. Acoustic Phonetics of /u/ in Earlier Baltimore English

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1. /u/ in Baltimore

Facts:

✧ Variation in /u/-advancement is widespread in varieties of English (Labov et al. 2006)
✧ Fronted /u/ is now a stereotypical feature of Baltimore English, shared with Philadelphia in the Mid-Atlantic region of the US (Labov et al. 2006)

However:

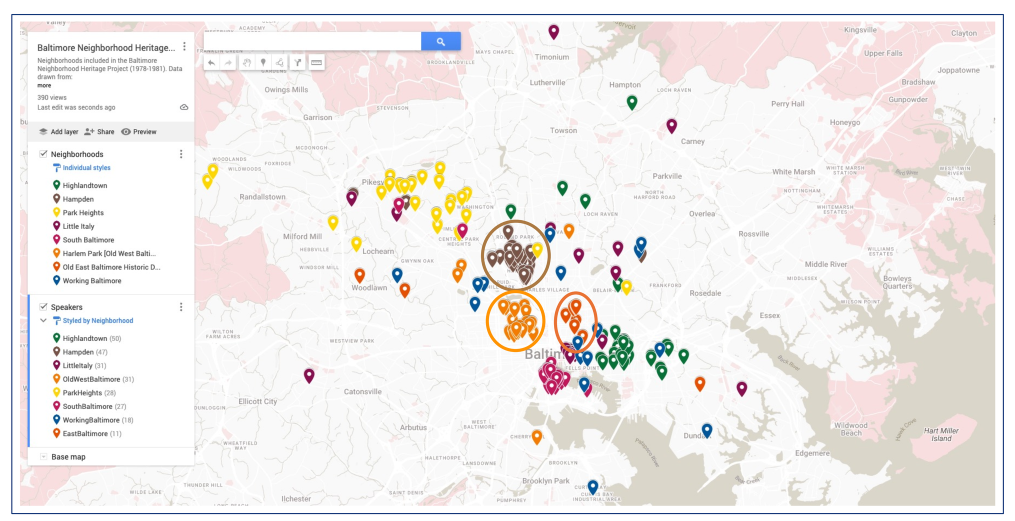
✧ <u>Change over time:</u> Earlier descriptions indicate /u/ was less fronted among Baltimore speakers born ca. 1900 (Kurath & McDavid 1961)
✧ <u>Ethnic variation:</u> Fronted /u/ now extends to Baltimore's large African American population (Jeremiah 2000), although /u/ remains backed in the typical African American Vowel Shift (Thomas 2007)
✧ <u>Vowel dynamics:</u> /u/ was more monophthongal among African American speakers born before World War I (Thomas 2007)

Research questions:

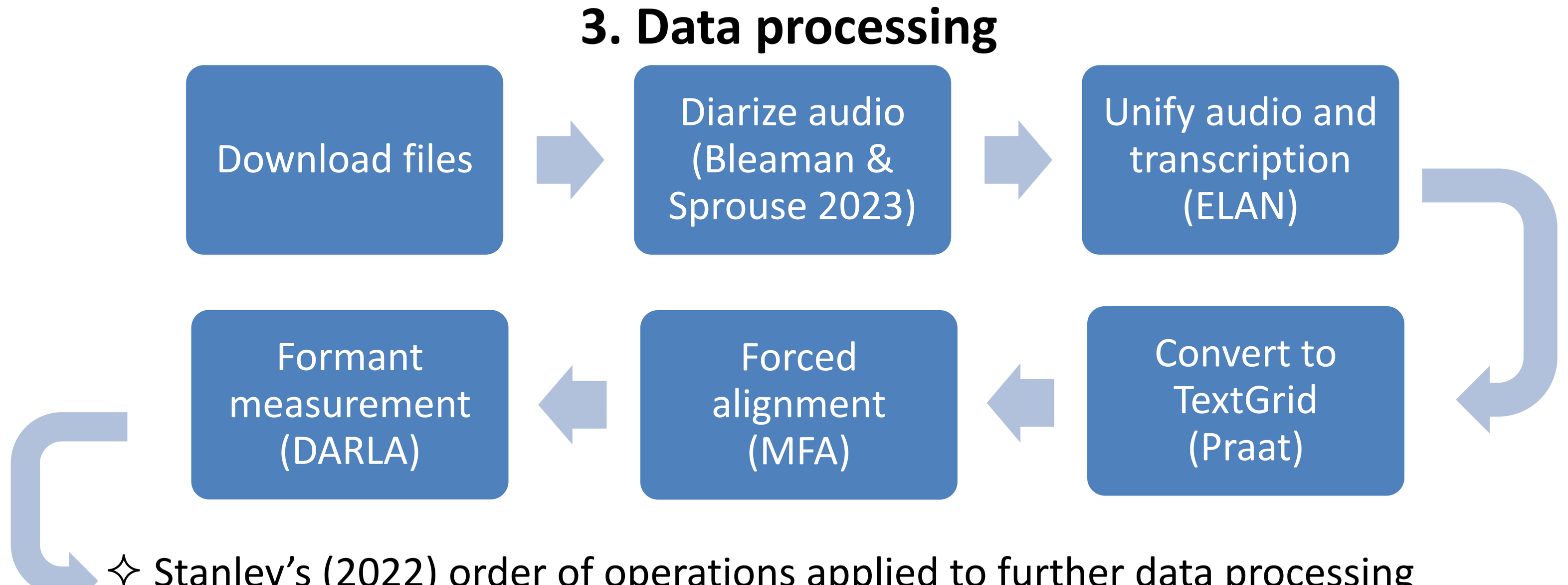
✧ What is the historic development of /u/-fronting in Baltimore, and how has the vowel's phonetic realization changed in apparent time?
✧ How do /u/'s position and vowel-inherent dynamics vary across contexts (TOOT, BOOT, POOL), neighborhoods, ethnicities and genders within Baltimore?

2. The Baltimore Neighborhood Heritage Project

✧ Community-based oral histories gathered 1978–1981
✧ Documented the history of 7 working-class and ethnic neighborhoods in Baltimore City, to democratize the historical record (Shopes 1981)
✧ 232 speakers (est. 320 hours audio), born 1879–1951 (median year of birth 1909)
✧ Many interviews manually transcribed, beginning in the 1980s
✧ Corpus digitized by the University of Baltimore's Special Collections & Archives



3. Data processing



✧ Stanley's (2022) order of operations applied to further data processing
  ✧ Logmeans normalization applied; custom stop words removed; stressed /u/ only

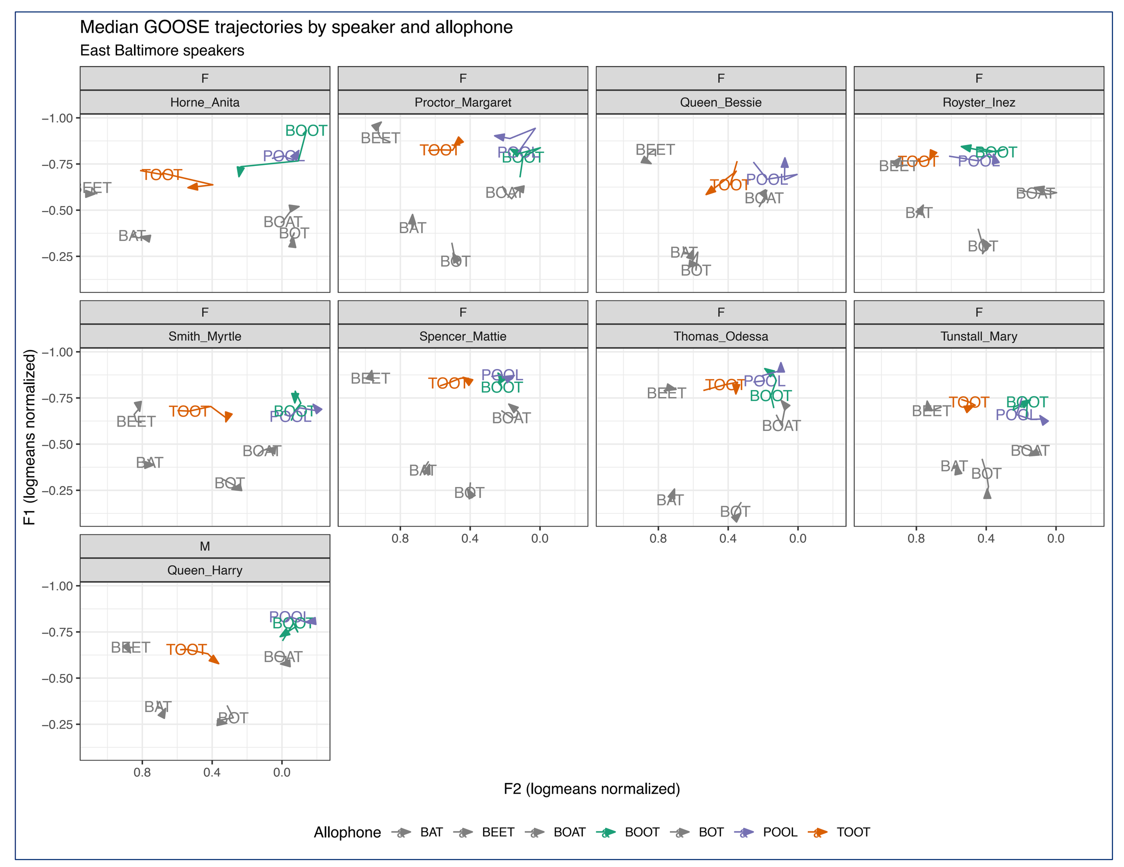
Speakers and available data from the Baltimore Neighborhood Heritage Project

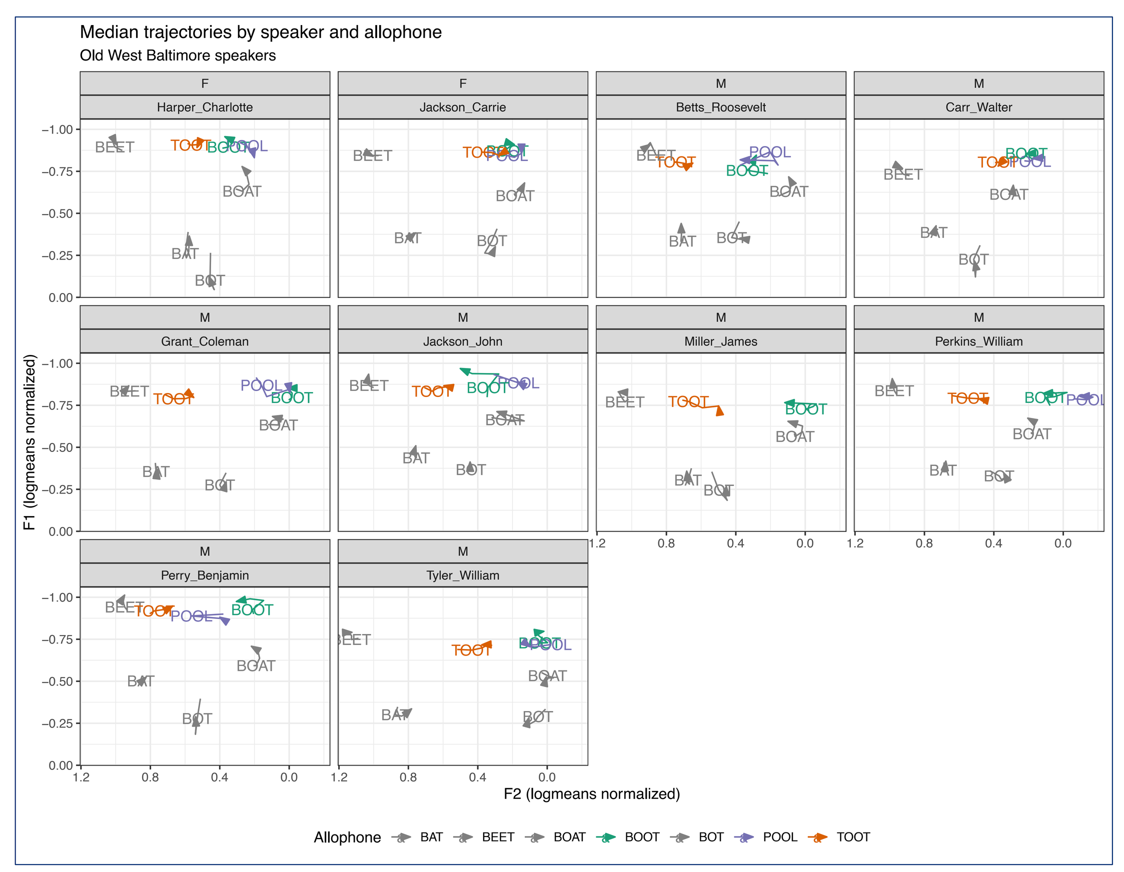
Neighborhood	Female	Male	Birth years	Ethnicity	BOOT	TOOT	POOL	Total
Old West Baltimore	2	8	1895-1940	Black	211	1584	148	1943
East Baltimore	8	1	1893-1918	Black	78	707	59	844
Hampden	7	4	1896-1935	White	185	1728	155	2068

4. Trajectories of /u/ in Baltimore

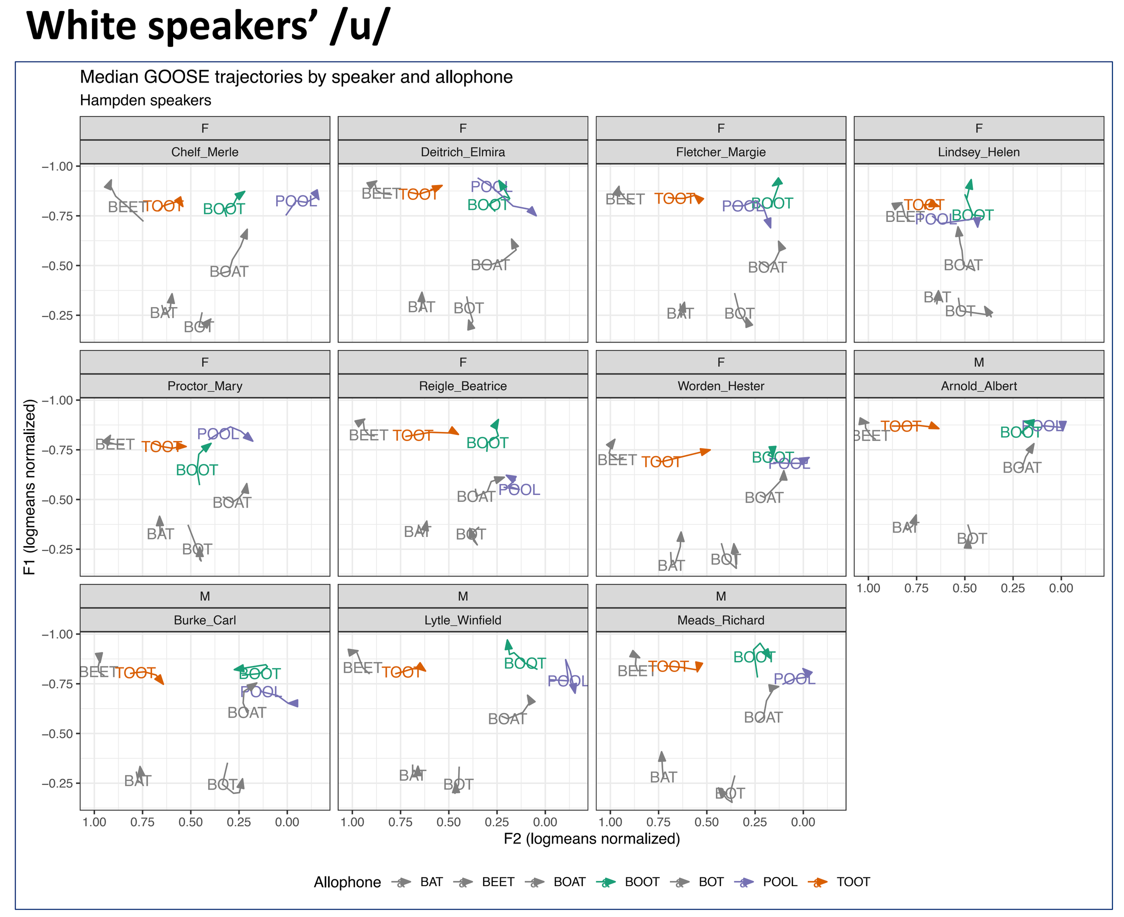
✧ Formant measurements across 5 time points illustrate dynamic trajectories of contextual /u/ allophones
✧ Significant variation occurs by allophone, neighborhood

Black speakers' /u/

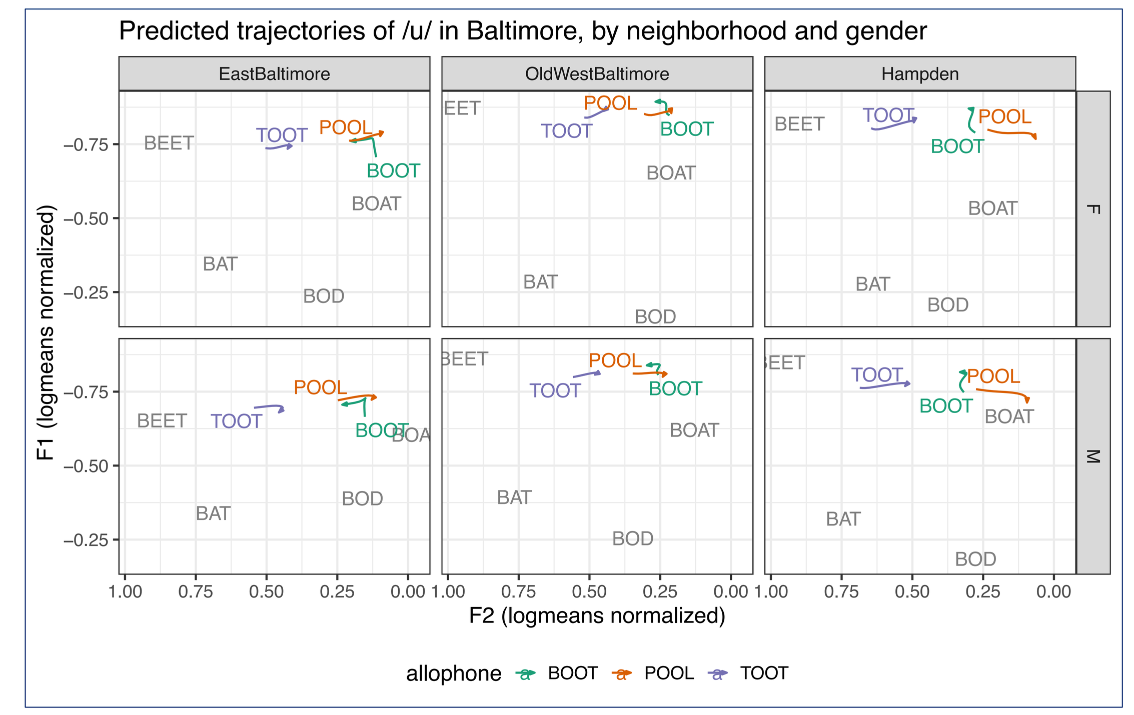


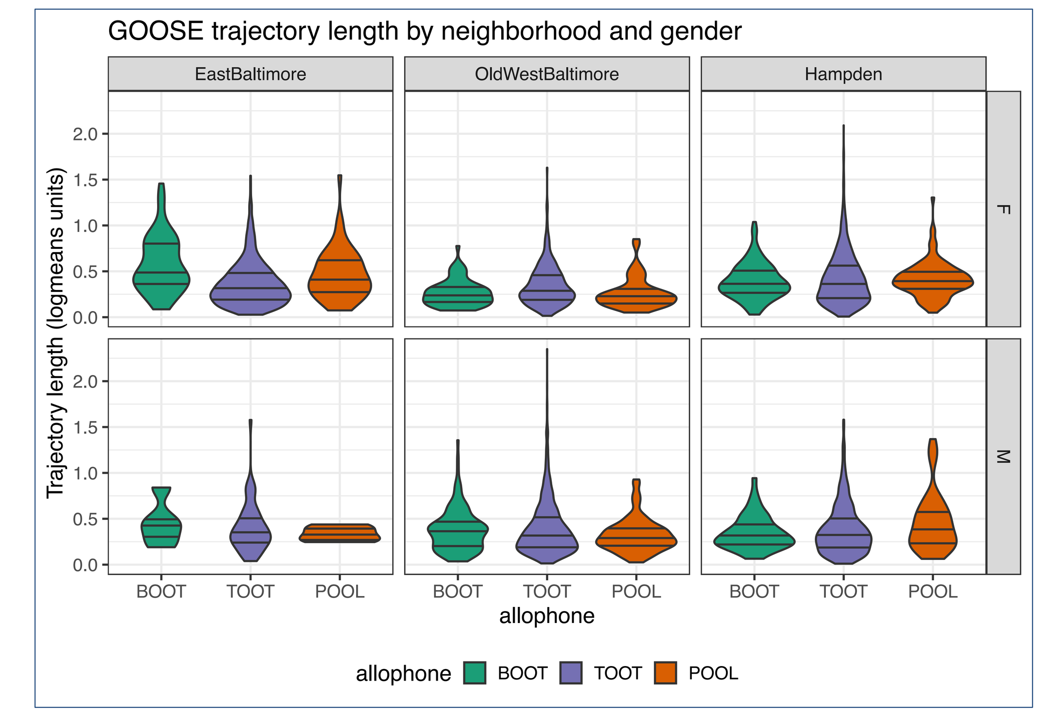


White speakers' /u/



5. Dynamics and advancement of /u/ across neighborhoods





✧ Generalized additive mixed-effects model (GAMM) fitted to trajectory data, summarizing across individuals
✧ Predicted trajectories show variation in position, shape

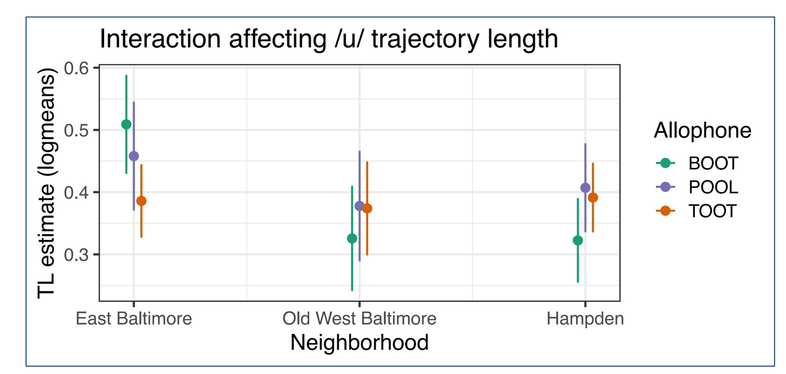
✧ Trajectory length (TL; Fox & Jacewicz 2009) evaluates extent of dynamic change in /u/'s formant values
✧ Results suggest differences across Black neighborhoods

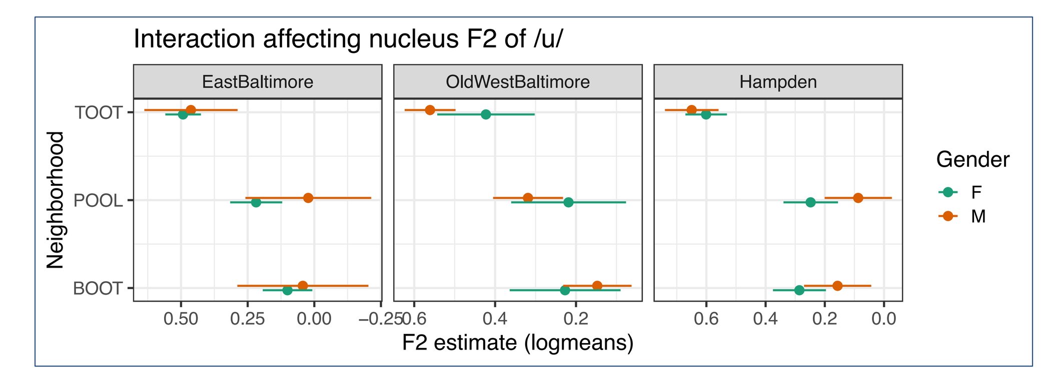
Statistical modeling

✧ Linear mixed-effects models applied to /u/ trajectory length and /u/ nucleus F2 values
✧ No significant effects or interactions found for year-of-birth. *model comparison with* `anova()`, p > *0.05*

```
TL: lmer(Y ~ gender + neighborhood*allophone + year-of-birth + duration + (1|word) + (1|speaker))
F2: lmer(Y ~ gender*neighborhood*allophone + year-of-birth + duration + (1|word) + (1|speaker))
```

✧ TL of non-postcoronal BOOT is significantly greater in East Baltimore than in Old West Baltimore ($\beta = 0.183$, $t(47.8) = 3.498$, $p < 0.01$) or Hampden ($\beta = 0.187$, $t(62.5) = 3.948$, $p < 0.001$). *pairwise comparisons,* `emmeans::pairs()`
✧ Nucleus F2 of /u/ varies by gender, allophone, and neighborhood. In East Baltimore vs. Hampden, females' BOOT is significantly fronter ($\beta = 0.185$, $t(3.566$, $p < 0.01)$; males' POOL is significantly backer ($\beta = -0.23$, $t(-3.703)$, $p < 0.01$)





6. Conclusions

✧ Greatest variation in /u/ occurs across phonological contexts: postcoronal TOOT is more fronted than BOOT or prelateral POOL
✧ Trajectory *length* varies across neighborhoods: Black speakers in East Baltimore may have a distinctively diphthongal BOOT
✧ *Advancement* of /u/ varies differently by allophone, neighborhood, and gender: due to tongue position, or lip rounding

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